

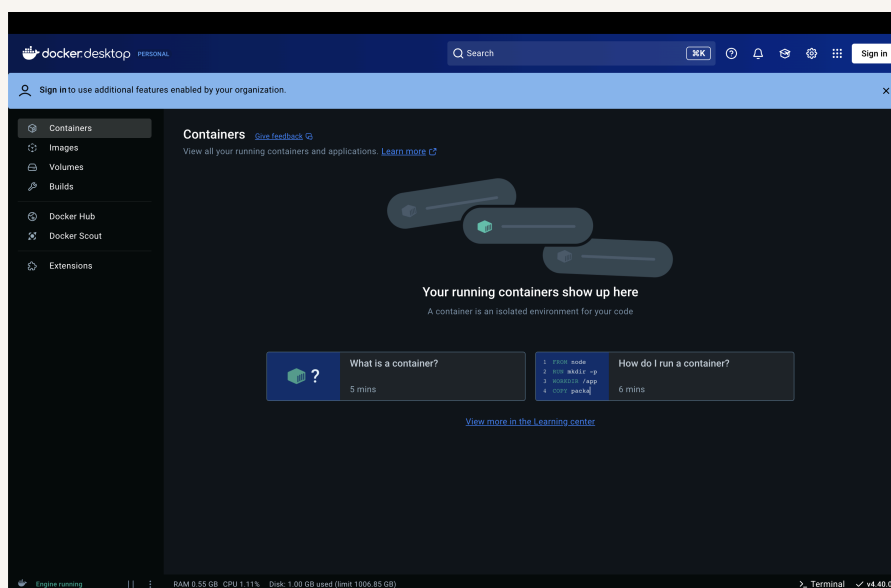


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Deploy an App with Docker



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Introducing Today's Project!

What is Docker?

Docker is an an open-source platform that automates the deployment of software applications by packaging them in containers. I used docker to create a custom image and then deployed it to Elastic Beanstalk

One thing I didn't expect...

How easy it is to deploy a custom image

This project took me...

Around an hour and a half



Understanding Containers and Docker

Containers

Containers help by packaging your application and everything it needs to run (i.e. dependencies) in one file. They also let you run other developers' applications and software much faster, since a container comes with everything you need.

A container image is a blueprint or template for containers. It gives Docker instructions on what to include in a container, such as application code, libraries, dependencies, and necessary files.

Docker

Docker is a tool that helps you create, manage, and deploy containers efficiently. Docker Desktop is an app that makes using Docker more interactive and efficient.

The Docker daemon is a background process that manages the Docker containers on your computer. It takes commands from the Docker client (i.e. commands you type into the terminal, or clicks you make through Desktop) and does the heavy lifting.



Running an Nginx Image

Nginx is a web server, which means it's a program that serves web pages to people on the internet.

The command I ran to start a new container was `docker run -d -p 80:80 nginx`.

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

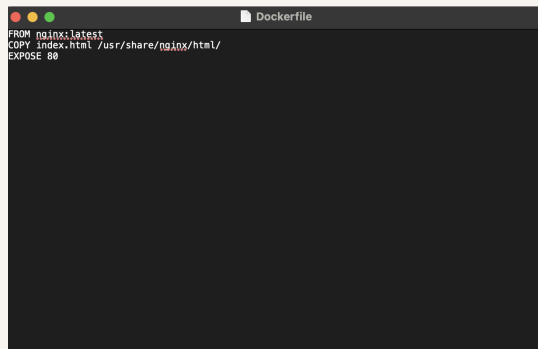


Creating a Custom Image

The Dockerfile is a document with all the instructions for building your Docker image. Docker would read a Dockerfile to understand how to set up your application's environment and which software packages it should install.

My Dockerfile tells Docker three things: - FROM nginx:latest means our image starts as a copy of the latest Nginx image - COPY index.html /usr/share/nginx/html/ replaces the default html file - EXPOSE 80 to tell the container to receive on port 80

The command I used to build a custom image with my Dockerfile was `docker build -t my-web-app .` The '.' at the end of the command means that Docker should find the Dockerfile in the current directory

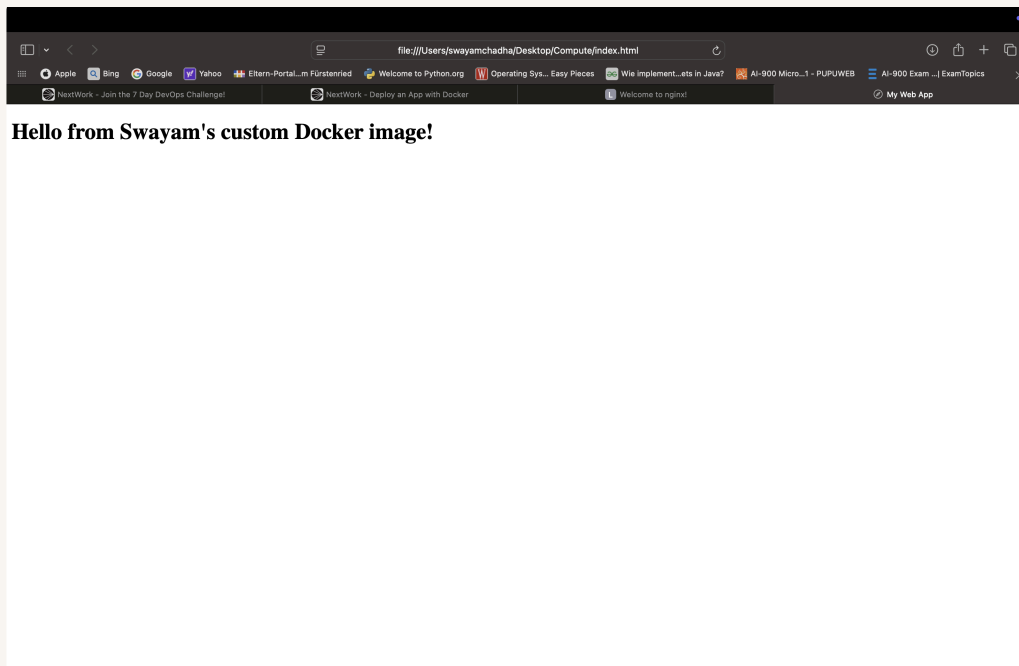


```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/
EXPOSE 80
```

Running My Custom Image

There was an error when I ran my custom image because my previous Nginx container was already running on port 80. I resolved this by stopping the container and running my-web-app again.

In this example, the container image is my-web-app made from the index.html and the container is the tool that runs the image.

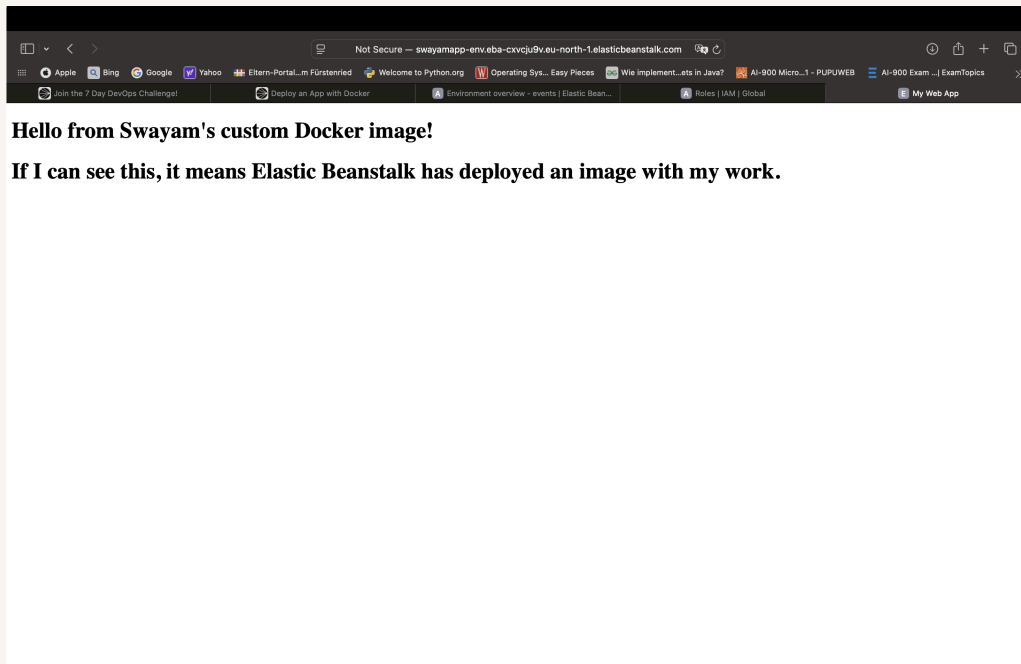




Elastic Beanstalk

Elastic Beanstalk is a service that makes it easy to deploy cloud applications without worrying about the underlying infrastructure.

Deploying my custom image with Elastic Beanstalk took me around an hour





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